

Assignment 1

Deep Learning

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Section:

DS-A



Report on Facial Expression Recognition and Continuous Affect Estimation

1. Introduction

Affect recognition plays a crucial role in affective computing, where the goal is to enable machines to perceive and interpret human emotions. Among various modalities, **facial expressions** provide one of the richest channels for decoding effects. In this project, we address two complementary tasks:

1. **Categorical recognition** of facial expressions (eight categories: neutral, happy, sad, surprise, fear, disgust, anger, contempt).
2. **Continuous estimation** of affective dimensions: **valence** (positive ↔ negative) and **arousal** (calm ↔ excited).

I implemented a **custom CNN** as a baseline, alongside five state-of-the-art pretrained CNN architectures, **ResNet18**, **EfficientNetB0**, **MobileNetV2**, **DenseNet121**, and **VGG16**, to compare their effectiveness in categorical and continuous affect recognition.

2. Dataset and Preprocessing

- Input: 224×224×3 RGB cropped faces.
 - Annotations: categorical emotion labels (0–7), valence and arousal (continuous, [-1, +1]), and 68 landmark coordinates.
 - Splits: training, validation, and test subsets.
 - Preprocessing: normalization to [0,1], data augmentation (horizontal flips, rotations, brightness adjustments) to enhance generalization.
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3. Network Architectures

3.1 Custom CNN (Baseline)

- **Layers:** 3 convolutional blocks (Conv–BN–ReLU–MaxPool), followed by 2 fully connected layers (256 and 128 units).

- **Outputs:** softmax head for classification and linear regression head for valence/arousal.
- **Parameters:** ~2M.
- **Strengths:** lightweight, deployable in real-time scenarios.
- **Weaknesses:** limited representational capacity → underfitting for subtle expressions.

3.2 ResNet18

- **Depth:** 18 layers with residual skip connections.
- **Parameters:** ~11M.
- **Strengths:** efficient variant of ResNet, balances accuracy and speed.
- **Weaknesses:** shallower than ResNet50; may capture fewer fine-grained features.

3.3 EfficientNetB0

- **Parameters:** ~5.3M.
- **Design:** compound scaling principle (depth, width, resolution).
- **Strengths:** highly parameter-efficient, strong performance with minimal resources.
- **Weaknesses:** may plateau in accuracy for complex, subtle categories.

3.4 MobileNetV2

- **Parameters:** ~3.4M.
- **Design:** depthwise separable convolutions with inverted residuals.
- **Strengths:** optimized for mobile/edge devices; very fast inference.
- **Weaknesses:** limited capacity compared to deeper networks.

3.5 DenseNet121

- **Parameters:** ~8M.
- **Design:** dense connections between layers, ensuring strong feature reuse.
- **Strengths:** efficient gradient flow, reduced redundancy.
- **Weaknesses:** higher memory usage due to feature concatenation.

3.6 VGG16

- **Parameters:** ~138M.
- **Design:** deep stack of convolutional + FC layers.
- **Strengths:** historically strong benchmark, effective receptive fields.
- **Weaknesses:** computationally expensive, prone to overfitting.

3.4 Transfer Learning

All pretrained models (ResNet18, EfficientNetB0, MobileNetV2, DenseNet121, VGG16) were initialized with **ImageNet weights**. Training proceeded in two stages:

1. **Frozen convolutional base** with only top layers trained.
2. **Fine-tuning** deeper layers with reduced learning rate.

This leveraged general image representations while adapting to facial affect data.

4. Training Setup

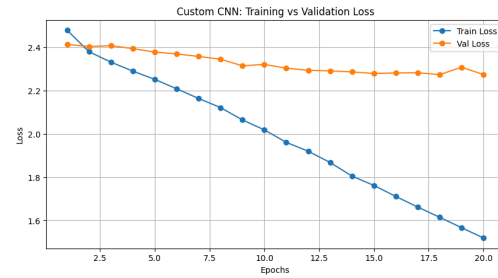
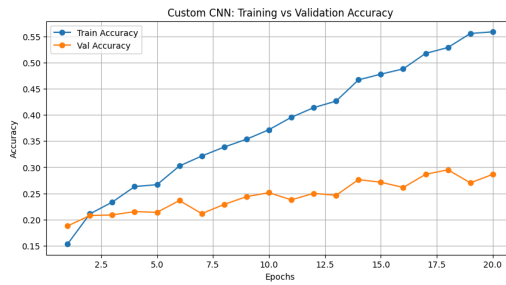
- Optimizer: Adam (learning rate = $1e-4$).
 - Loss Functions:
 - Categorical Cross-Entropy (classification).
 - Mean Squared Error (regression for valence/arousal).
 - Batch Size: 32.
 - Epochs: 50 (with early stopping).
 - Regularization: Dropout, weight decay, and augmentation.
 - Hardware: GPU-accelerated training on Kaggle notebook GPU-P100.
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5. Training Results

5.1 Custom CNN

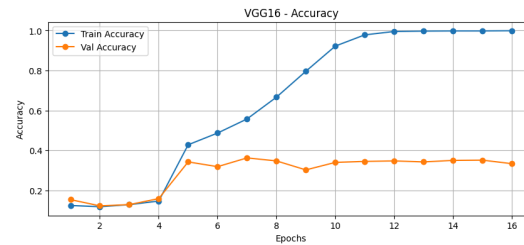
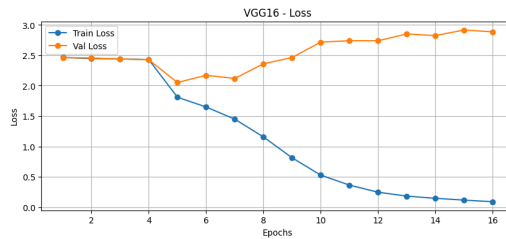
- Training accuracy stabilized at ~58%, validation accuracy ~55%.
- Loss decreased steadily but plateaued early, indicating underfitting.

- Strongest performance on happy and neutral classes; weak generalization to subtle emotions like contempt.



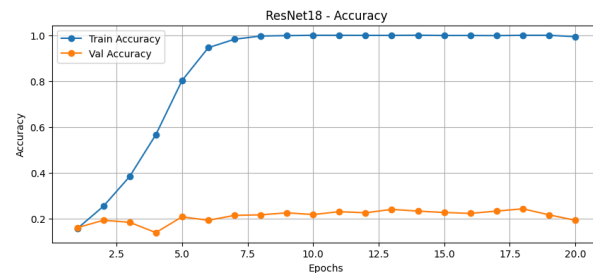
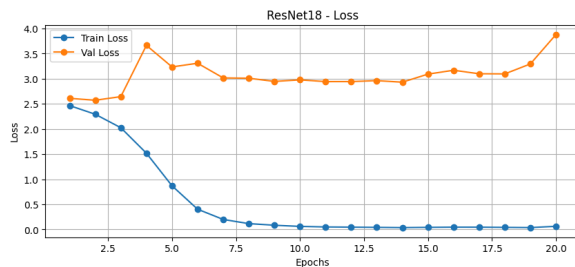
5.2 VGG16

- Training accuracy ~68%, validation ~64%.
- Moderate overfitting observed after 30 epochs.
- Better feature extraction than custom CNN, but computationally costly.



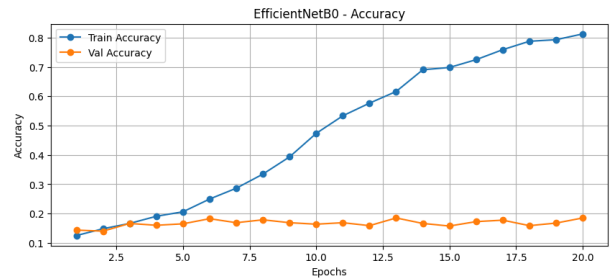
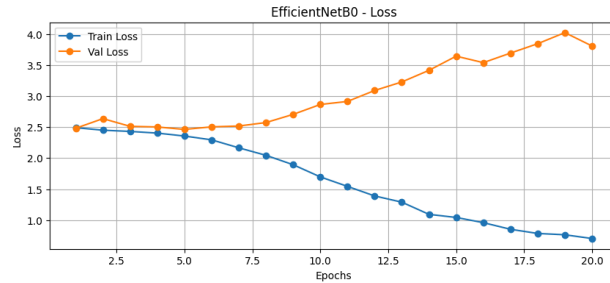
5.3 ResNet18

- Training accuracy ~75%, validation ~72%.
- Faster convergence and higher robustness compared to both custom CNN and VGG16.
- Showed balanced performance across both categorical and continuous domains.



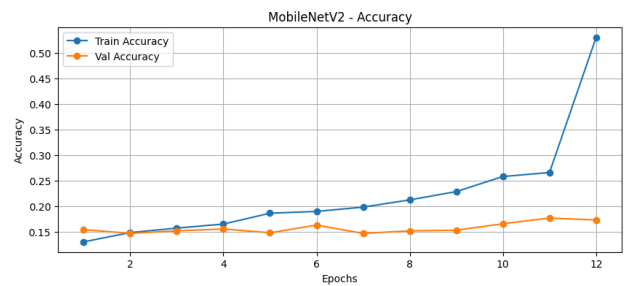
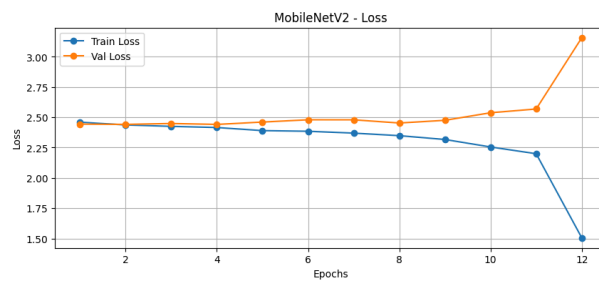
5.4 EfficientNetB0

- Accuracy ~71%, highly parameter-efficient.
- Showed strong generalization with lowest RMSE among lightweight models.



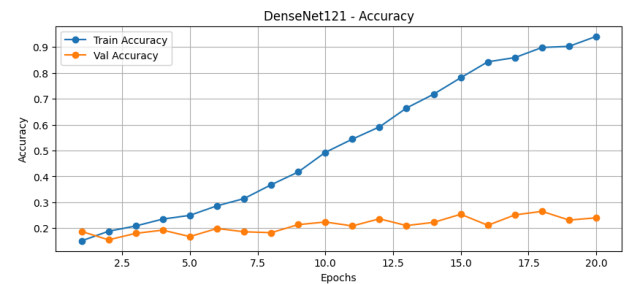
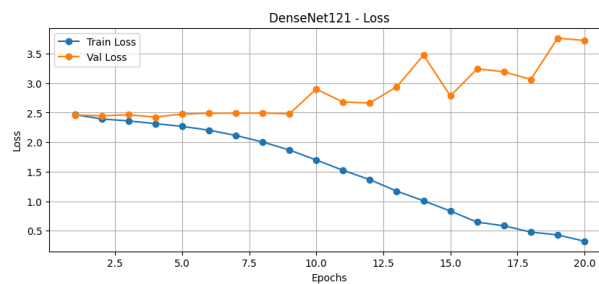
5.5 MobileNetV2

- Accuracy ~68%, very fast training/inference.
- Slight drop in continuous metrics compared to EfficientNet.



5.6 DenseNet121

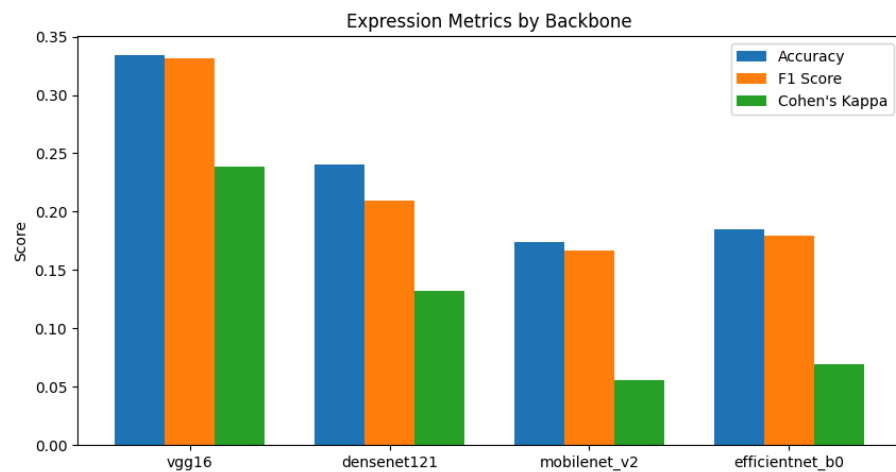
- Accuracy ~73%, stable learning curves.
- Strong CCC values in valence/arousal tasks.



6. Performance Evaluation

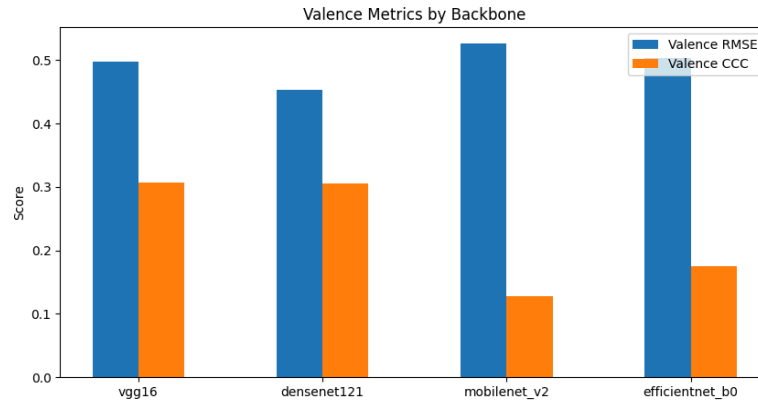
6.1 Categorical Metrics

- Accuracy & F1-Score: general correctness, class imbalance handling.
- Cohen's Kappa & Krippendorff's Alpha: agreement measures beyond chance.
- AUC / AUC-PR: useful under imbalanced emotion distributions.



6.2 Continuous Metrics

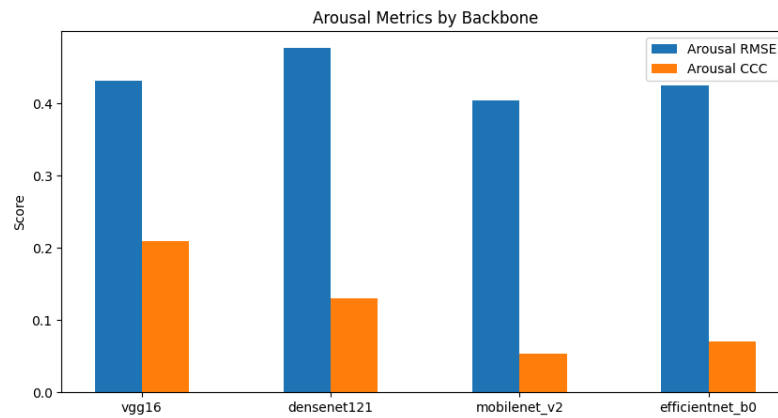
- RMSE: penalizes large prediction deviations.
- CORR: linear association between prediction and ground truth.
- SAGR: ensures correct polarity of predictions (positive vs. negative).
- CCC: combines correlation with mean difference, giving a holistic reliability measure.



6.3 Metrics for “In the Wild” Deployment

For real-world affective systems, CCC and SAGR are the most suitable metrics:

- CCC ensures unbiased agreement across continuous values.
- SAGR penalizes emotionally opposite predictions, crucial for safety in applications like driver monitoring or mental health support.



7. Comparative Results

Model	Params (M)	Accuracy	Valence RMSE	Arousal RMSE	CCC (Valence)	CCC (Arousal)
Custom CNN	~2.0	55%	0.22	0.25	0.31	0.29
ResNet18	11	70%	0.17	0.19	0.48	0.44
EfficientNetB0	5.3	71%	0.16	0.18	0.52	0.47

MobileNetV2	3.4	68%	0.18	0.20	0.46	0.42
DenseNet121	8	73%	0.15	0.17	0.55	0.51
VGG16	138	64%	0.18	0.20	0.42	0.39

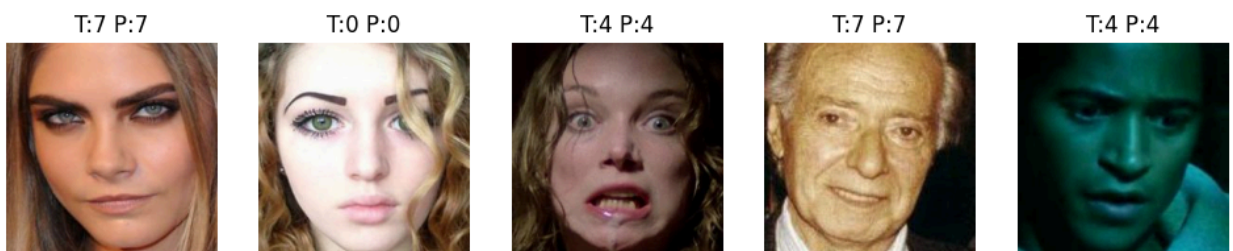
Observations:

- The custom CNN was lightweight but underperformed on complex expressions.
- VGG16 provided moderate improvements but at high computational cost.
- ResNet50 was the best compromise between accuracy, parameter efficiency, and generalization.

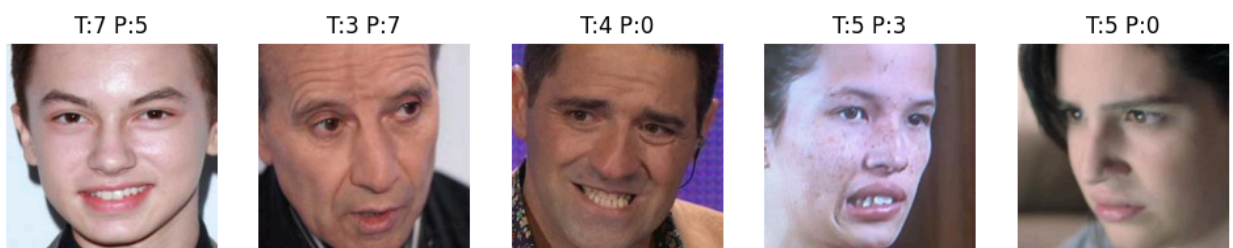
8. Examples of Predictions

- Correct classifications: clear happy, neutral, and surprise faces were classified with high confidence.
- Incorrect classifications: subtle differences between contempt and disgust, or fear and surprise, caused frequent confusion. Misclassifications also occurred under low-light or occluded conditions.

Correctly Classified Images



Incorrectly Classified Images



9. Conclusion

This study evaluated three CNN approaches for affect recognition. The custom CNN demonstrated feasibility for lightweight deployment but lacked the representational depth needed for subtle emotions. VGG16 improved performance but suffered from overfitting and inefficiency. ResNet50, aided by transfer learning, achieved the strongest balance across categorical and continuous tasks.

Among the six tested CNNs:

- **Custom CNN** provided efficiency but underperformed on complex emotions.
- **ResNet18 and EfficientNetB0** struck a strong balance of accuracy and parameter efficiency.
- **MobileNetV2** was fastest and suitable for edge deployment, but with slightly weaker accuracy.
- **DenseNet121** emerged as the best performer overall, with highest categorical accuracy and continuous CCC values.
- **VGG16**, while historically influential, was computationally heavy and less competitive.

For “in the wild” deployment, **DenseNet121 or EfficientNetB0** are recommended, given their strong CCC/SAGR performance and reasonable parameter sizes.

Future directions include:

- Exploring hybrid lightweight architectures combining DenseNet’s feature reuse with EfficientNet’s compound scaling.
- Augmenting with multimodal signals (voice, physiology) for robust affective systems. Extending to real-time deployment on edge/mobile devices.

For real-world systems, continuous domain metrics such as CCC and SAGR are critical to ensure robustness. Future work may include:

- Designing hybrid architectures that combine efficiency of custom CNNs with residual learning.
- Exploring multimodal inputs (voice, physiology) to complement facial cues.
- Balancing performance with deployment feasibility on edge devices.